Introduction to Database Systems (CSD202)

Lab Assignment#7

Date: 16/03/18

You have to upload the screen shot (black screen) of output with the queries one by one, in a word doc & file name must be->Roll No _Assignment#7.

Procedure: -

1. Use DBMS2019 for all problems.

```
mysql> use dbms2019;
Database changed
```

2. Create a procedure **count_rcrd** to count the number of records of any table of selected database (for example actor).

```
mysql> create procedure count_rcrd (IN customer varchar(30))
    -> begin
    -> set @table = customer;
    -> set @sql_text = concat('select count(*) from ',@table);
    -> prepare stmt from @sql_text;
    -> execute stmt;
    -> end;
    -> //
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> call count_rcrd('customer');
+----------+
| count(*) |
+----------+
|        8 |
+----------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.02 sec)
```

3. Create a Stored Procedure "**CreateTable**" to create a dynamic table (means you will pass table name at the time of procedure call) as per given below declaration.

Attribute(Field/Column) Name	Data Declaration
S_ID	Int
Name	Varchar(30)
Department	Varchar(30)
DBMS_Marks	Decimal(5,2)

- For now, pass table name **student**

```
mysql> create procedure CreateTable (IN tablename varchar(30))
    -> begin
    -> set @table = tablename;
    -> set @sql_text = concat('create table ',@table,'(s_id int, name varchar(30), department varchar(30), DBMS_Marks
 decimal(5,2))');
    -> prepare stmt from @sql_text;
    -> execute stmt;
    -> end
    -> //
Query OK, 0 rows affected (0.00 sec)

mysql> call CreateTable("Student");
    -> //
Query OK, 0 rows affected (0.16 sec)

mysql> describe student;
    -> //
+------------+--------------+------+-----+---------+-------+
| Field      | Type         | Null | Key | Default | Extra |
+------------+--------------+------+-----+---------+-------+
| s_id       | int(11)      | YES  |     | NULL    |       |
| name       | varchar(30)  | YES  |     | NULL    |       |
| department | varchar(30)  | YES  |     | NULL    |       |
| DBMS_Marks | decimal(5,2) | YES  |     | NULL    |       |
+------------+--------------+------+-----+---------+-------+
4 rows in set (0.01 sec)
```

4. Create a procedure **Insert_record** to insert below mention data to the table.

S_ID	Name	Department	DBMS_Marks
1001	Arush Kharbanda	CSE	74.5
1002	Ayush Malik	Mechinical	82.0
1003	Palak Nanda	Civil	94.5
1004	Abhishek Chadha	CSE	48.5
1005	Abhyudaya Sharma	Electrical	70.5
1006	Akarsh Kala	Mechinical	68.0
1007	Alaparthy Meghana	Civil	95.0
1008	Ambreesh	CSE	56.0
1009	Anish R	Electrical	65.5
1010	Ankita Durgavajula	Mechinical	35.0

```
mysql> create procedure insert_record(IN id int, IN sname varchar(30), IN dept varchar(30), IN marks decimal(5,2))
    -> begin
    -> insert into student values (id,sname,dept,marks);
    -> end
    -> //
Query OK, 0 rows affected (0.00 sec)

mysql> call insert_record (1001,'Arush Kharbanda','CSE',74.5)//
Query OK, 1 row affected (0.03 sec)

mysql> call insert_record (1002,'Ayush Malik','Mechanical',82.0)//
Query OK, 1 row affected (0.03 sec)

mysql> call insert_record (1003,'Palak Nanda','Civil',94.5)//
Query OK, 1 row affected (0.01 sec)

mysql> call insert_record (1004,'Abhishek Chadha','CSE',48.5)//
Query OK, 1 row affected (0.03 sec)

mysql> call insert_record (1005,'Abhyudaya Sharma','Electrical',70.5)//
Query OK, 1 row affected (0.02 sec)

mysql> call insert_record (1006,'Akarsh Kala','Mechanical',68)//
Query OK, 1 row affected (0.23 sec)

mysql> call insert_record (1007,'Alaparthy Meghana','Civil',95)//
Query OK, 1 row affected (0.03 sec)

mysql> call insert_record (1008,'Ambreesh','CSE',56)//
Query OK, 1 row affected (0.03 sec)

mysql> call insert_record (1009,'Anish R','Electrical',65.5)//
Query OK, 1 row affected (0.03 sec)

mysql> call insert_record (1010,'Ankita Durgavajula','Mechanical',35)//
Query OK, 1 row affected (0.16 sec)
```

5. Display table student.

```
mysql> select * from student//
+------+--------------------+------------+------------+
| s_id | name               | department | DBMS_Marks |
+------+--------------------+------------+------------+
| 1001 | Arush Kharbanda    | CSE        |      74.50 |
| 1002 | Ayush Malik        | Mechanical |      82.00 |
| 1003 | Palak Nanda        | Civil      |      94.50 |
| 1004 | Abhishek Chadha    | CSE        |      48.50 |
| 1005 | Abhyudaya Sharma   | Electrical |      70.50 |
| 1006 | Akarsh Kala        | Mechanical |      68.00 |
| 1007 | Alaparthy Meghana  | Civil      |      95.00 |
| 1008 | Ambreesh           | CSE        |      56.00 |
| 1009 | Anish R            | Electrical |      65.50 |
| 1010 | Ankita Durgavajula | Mechanical |      35.00 |
+------+--------------------+------------+------------+
10 rows in set (0.00 sec)
```

6. Create & Call a Stored Procedure "**display_table"** to display the data of any table (you just need to pass the table name) at the time of procedure call.

```
mysql> create procedure display_table(IN tablename varchar(30))
    -> begin
    -> set @table = tablename;
    -> set @sql_text = concat('select * from ',@table);
    -> prepare stmt from @sql_text;
    -> execute stmt;
    -> end
    -> //
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> call display_table('student')//
+------+--------------------+------------+------------+
| s_id | name               | department | DBMS_Marks |
+------+--------------------+------------+------------+
| 1001 | Arush Kharbanda    | CSE        |      74.50 |
| 1002 | Ayush Malik        | Mechanical |      82.00 |
| 1003 | Palak Nanda        | Civil      |      94.50 |
| 1004 | Abhishek Chadha    | CSE        |      48.50 |
| 1005 | Abhyudaya Sharma   | Electrical |      70.50 |
| 1006 | Akarsh Kala        | Mechanical |      68.00 |
| 1007 | Alaparthy Meghana  | Civil      |      95.00 |
| 1008 | Ambreesh           | CSE        |      56.00 |
| 1009 | Anish R            | Electrical |      65.50 |
| 1010 | Ankita Durgavajula | Mechanical |      35.00 |
+------+--------------------+------------+------------+
10 rows in set (0.00 sec)

Query OK, 0 rows affected (0.05 sec)
```

7. Create & Call a Stored Procedure "**Update_Record"** to update name of a student for example, set the student name "Ram" whose S_ID is 1006 at the time of procedure call.

```
mysql> create procedure update_record(IN name varchar(30), IN id int)
    -> begin
    -> update student set Name = name where s_id = id;
    -> end
    -> //
Query OK, 0 rows affected (0.00 sec)

mysql> call update_record ('Ram',1006);
    -> //
Query OK, 1 row affected (0.01 sec)
```

8. Display the definition of **update_record** procedure.

```
mysql> show create procedure update_record//
+---------------+----------------------------------------------------------------+----------------------------------
-----------------------------------------------------------------------------------------------------------------+
---------------------+----------------------+--------------------+
| Procedure     | sql_mode                                                       | Create Procedure
                                                                                                                 |
 character_set_client | collation_connection | Database Collation |
+---------------+----------------------------------------------------------------+----------------------------------
-----------------------------------------------------------------------------------------------------------------+
---------------------+----------------------+--------------------+
| update_record | STRICT_TRANS_TABLES,NO_AUTO_CREATE_USER,NO_ENGINE_SUBSTITUTION | CREATE DEFINER=`root`@`localhost`
PROCEDURE `update_record`(IN name varchar(30), IN id int)
begin
update student set Name = name where s_id = id;
end | utf8                 | utf8_general_ci      | utf8_general_ci    |
+---------------+----------------------------------------------------------------+----------------------------------
-----------------------------------------------------------------------------------------------------------------+
---------------------+----------------------+--------------------+
1 row in set (0.00 sec)
```

9. Drop all the procedures you have created recently.

```
mysql> drop procedure CreateTable;
    -> //
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> drop procedure count_rcrd//
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> drop procedure insert_record//
Query OK, 0 rows affected (0.00 sec)

mysql> drop procedure update_record//
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> drop procedure display_table//
Query OK, 0 rows affected (0.00 sec)
```